

Chapter 44: In Silico Evaluation of *L*-theanine, a component of tea, as a potential MAO-B Inhibitor for Dopamine Preservation in Parkinson's Disease

Satarupa Deb^{1*}, Venkateshwarla Rama Raju², Nisat Firduse¹, Dipta Rajkumari

¹Department of Zoology, Patharkandi College, Patharkandi, Sribhumi, Assam, India.

²Jawaharlal Nehru Technological University (JNTU), Hyderabad, Telangana, India.

Abstract

Monoamine oxidase B (MAO-B) plays a pivotal role in the oxidative deamination of dopamine, a process crucial for maintaining neurotransmitter homeostasis in the central nervous system. In Parkinson's disease (PD), increased MAO-B activity exacerbates dopamine depletion, contributing to disease progression. MAO-B inhibitors such as selegiline, rasagiline, and safinamide are widely used to mitigate dopamine breakdown, but their long-term efficacy and side effects necessitate the search for alternative therapeutic agents. *L*-theanine, a bioactive amino acid found in green tea (*Camellia sinensis*), has demonstrated neuroprotective properties and the ability to modulate neurotransmitter levels. However, its interaction with MAO-B remains largely unexplored. This study investigates the potential of *L*-theanine as a MAO-B inhibitor through molecular docking analyses, comparing its binding affinity with established inhibitors. Additionally, the drug-likeness properties of *L*-theanine are assessed to determine its viability as a therapeutic lead compound. Our findings provide insights into the molecular interactions between *L*-theanine and MAO-B, supporting its potential as a natural therapeutic agent for PD and other neurodegenerative disorders characterized by dopamine dysregulation.

Keywords: Monoamine oxidase B (MAO-B), Parkinson's disease (PD), dopamine metabolism, MAO-B inhibitors, *L*-theanine, molecular docking.

1. Introduction

Monoamine oxidase B (MAO-B) is a flavin-containing enzyme predominantly expressed in the brain that regulates dopamine homeostasis by catalyzing its oxidative deamination to 3,4-dihydroxyphenylacetic acid (DOPAC) (Bortolato et al., 2008; Edmondson & Binda, 2018; Tan et al., 2022). Together with monoamine oxidase A (MAO-A), MAO-B belongs to the monoamine oxidase family, although the two isoforms differ in tissue distribution and substrate specificity (Chew et al., 2023; Harro & Orelund, 2016). In the central nervous system, particularly within the striatum, MAO-B plays a major role in dopamine metabolism, thereby regulating neuronal function and motor behaviour (Cho et al., 2021; Jaisa-aad et al., 2024).

The pathological significance of MAO-B has been extensively investigated in Parkinson's disease (PD), a neurodegenerative disorder characterized by progressive degeneration of dopaminergic neurons in the substantia nigra and consequent dopamine depletion, leading to motor impairments such as tremor, bradykinesia, and rigidity (Marino et al., 2020; Regensburger et al., 2023; Vijiaratnam et al., 2021). Elevated MAO-B activity further accelerates dopamine degradation, thereby exacerbating disease progression (Kumar & Andersen, 2004). Consequently, selective MAO-B inhibition has emerged as an established therapeutic strategy to preserve dopamine levels and improve motor function in PD patients (Jost, 2022).

Several selective MAO-B inhibitors, including selegiline, rasagiline, and safinamide, are currently used in clinical practice (Müller & Möhr, 2019). These agents inhibit dopamine metabolism by binding to

the enzyme's active site, thereby alleviating motor symptoms (Finberg, 2018). However, their long-term use is associated with adverse effects, drug interactions, and variable therapeutic efficacy, underscoring the need for safer and more effective alternatives (Riederer & Müller, 2018; Szökö et al., 2018; Tsuboi et al., 2022; Yamada & Yasuhara, 2004).

Natural products have attracted increasing attention as potential neuroprotective agents because of their favourable safety profiles and multiple biological activities (Deb et al., 2019; Grabska-Kobylecka et al., 2023; Kuboyama et al., 2014). L-theanine, a non-proteinogenic amino acid abundant in green tea (*Camellia sinensis*), readily crosses the blood-brain barrier and modulates neurotransmission, stress responses, and neurotransmitter homeostasis (Chu et al., 1999; Dramard et al., 2018; Nathan et al., 2006; Yoto et al., 2012). Emerging evidence suggests that L-theanine may also interact with MAO-B, although its inhibitory potential and molecular binding mechanism remain poorly understood (Shao et al., 2022).

Therefore, the present study employed molecular docking to investigate the interaction of L-theanine with MAO-B and compared its binding affinity with established MAO-B inhibitors, including rasagiline and selegiline. In addition, the drug-likeness properties of L-theanine were evaluated to assess its potential as a lead compound. The findings may provide mechanistic insights into the neuroprotective effects of L-theanine and support its future development as a natural therapeutic candidate for Parkinson's disease and related neurodegenerative disorders.

2. Materials and methods

2.1. Receptors and Ligands

The three-dimensional structure of human MAO B enzyme (PDB id: 2V5Z) was downloaded from RCSB Protein Data Bank (www.rcsb.org/pdb) in PDB format. The structure of our ligand of interest, L-theanine, along with two other known competitive MAO-B receptor antagonists, safinamide, procarbazine, selegiline, methamphetamine, and rasagiline were downloaded in .mol format from CAS DataBase List of Chemical Book (<https://www.chemicalbook.com>). The selection of the structure was based on the receptor model, resolution, and source organism. The structure 2V5Z has 2 chains of equal length, determined by X-ray diffraction at a resolution of 1.60Å and has two bound ligands, flavin-adenine dinucleotide (FAD) and (S)-(+)-2-[4-(fluorobenzyloxy-benzylamino) propionamide] (SAG).

2.2. Stereochemical Quality Assessment of the Receptor

Ramachandran plots were generated for the receptor using PROCHECK 3.6.2 module available at the PDBSum server to assess the stereochemical quality of the receptors (Laskowski et al., 1993).

2.3. Detection of Available Cavities in the Receptor

For detecting the available cavities in the protein, which have the potential to allow the binding of the ligands, Molegro Virtual Docker 7.0.0 (MVD) was used (Thomsen & Christensen, 2006). The cavity volumes, the amino acids present and the cavity coordinates were determined. These pockets were confirmed and the druggability of these pockets was revealed by using Cavity Plus (<http://www.pkumdl.cn:8000/cavityplus/index.php>), which is an online tool for protein cavity detection (Xu et al., 2018).

2.4. Pharmacophore Generation of the Ligands

For visualisation and comparison of the potential interaction groups on the ligand used in the study, pharmacophores were generated using the CavPharmer module (<http://www.pkumdl.cn:8000/cavityplus/computation.php#pharmacophoreModule>) of the Cavity Plus online tool (Xu et al., 2018).

2.5. The Docking Simulation

The receptor (2V5Z) was loaded in Molegro Virtual Docker 7.0.0 software. Preparation of the protein and correction of amino acids for protonation, as well as energy minimisation, were carried out consecutively. Extra water molecules were removed, and pre-existing ligands in the receptor pockets (if any) were withdrawn. The final binding site was selected as the active pocket that binds with the ligands in the most suitable poses. The scoring function was selected to be MolDock Score with Grid Resolution set to 0.30 Å (Mazumder et al., 2013). The default binding coordinates for L-theanine in the MAO-B receptor were selected for docking. The results of molecular docking were visualized using a BIOVIA discovery studio visualizer (Baroroh, S.Si. et al., 2023). Ligand maps were generated, which showed the hydrogen bonding interactions, steric interactions and the electro-static interactions between the ligand and the receptors.

2.6. Assessment of drug-likeness properties of different ligands

The InChI structures of the compounds were retrieved from the PubChem compounds database and subsequently converted to the desired structure format using Open Babel for docking applications. Following this, the drug likeness properties of the compounds were assessed utilizing the Molsoft L.L.C. online server (Zhong et al., 2014).

3. Results

3.1. Preparation of Ligands and Target Receptors for Molecular Docking

3.1.1. Stereochemical Quality of selected MAO-B protein.

The Ramachandran plot generated for the MAO-B receptor indicates that 93.4% of its residues are situated within the most favoured regions, with only 5.9% falling into the additionally allowed regions of the plot (Fig. 1). Proteins with a majority of residues falling within the most favoured regions of the Ramachandran plot are considered to have a higher quality structure, indicating proper folding and stability. The Ramachandran plot analysis indicates that the selected receptor exhibits excellent stereochemical quality, rendering it suitable for conducting molecular binding studies with our target ligands.

3.1.2. Identification of cavities in the receptor.

The MAO-B protein receptor was analysed using the CavityPlus module, revealing the presence of 31 cavity sites within both chains of the protein structure. These cavities represent potential active sites where interactions with ligands can occur. From this pool, two cavities were singled out based on their exceptionally high druggability scores of 2698 and 2349. These selected cavities were prioritized for further investigation through specific ligand docking studies.

3.2. Docking Analysis of the target receptor and ligands

3.2.1. Interactions of MAO-B receptor with Sildenafil and Procarbazine

From the docking study, it can be interpreted that the compound safinamide binds to MAO-B receptor by forming hydrogen bonds with tyrosine 326(A), isoleucine 199(A) and tyrosine 398(A), while forming a carbon-hydrogen bond with Glycine 58(A) and another pi-pi stacked bond with tyrosine 398(A) residue. Steric interactions do not occur between the ligands and the receptor. A pi-sulphur bond can be identified in safinamide which is attached to cysteine 172(A) and a pi-alkyl bond is present with leucine 171(A)(Fig. 2 A). Apart from conventional hydrogen bonds in procarbazine among Glutamine 206(A), Methionine 436(A), Glycine 434(A), a carbon hydrogen bond is also formed at leucine 171(A). There is an additional pi-pi stacked bond with tyrosine 435(A) and a pi-alkyl bond with arginine 42(A). Methionine 436(A) has both conventional hydrogen bond as well as an alkyl bond (Fig. 2 B). Thus, it can be noted that conventional hydrogen bonds are equal in number in safinamide and procarbazine. However, the overall bonds are comparatively more in procarbazine.

3.2.2. Interactions of MAO-B receptor with Selegiline and R-Methamphetamine

Both the compounds selegiline and methamphetamine possess similar structures, which can be explained by the fact that selegiline is partially metabolized to produce N-desmethylselegiline, L-amphetamine, and L-methamphetamine. The structures of selegiline and methamphetamine are indeed similar, and this similarity arises from their shared chemical backbone, specifically their amine functional groups and substituted phenyl rings. This structural resemblance contributes to their pharmacological effects, as both compounds can interact with adrenergic and dopaminergic receptors in the brain, albeit with different affinities and mechanisms of action. However, the difference in number of bonds formed by these compounds with MAO-B receptor is notably high. Amphetamine showed the highest number of interacting bonds as compared to other ligands. Unlike other compounds, selegiline has no hydrogen bonds with the receptor (Fig. 3 A). It only exhibits pi-pi stacked, pi-alkyl and Carbon-Hydrogen bonds. On the other hand, methamphetamine exhibits multiple interactions with the MAO-B receptor, the highest of which are the weak interactions of van der Waal formed by Isoleucine (B)272, Histidine (A)273, Valine (B)235, Proline (B)234, Valine (B)10, Glutamic acid (B)34, Glycine (B)12, Arginine (B)36, and Glycine (B)11. Apart from these, it also displays alkyl bonds with Alanine (B)35, Lysine (B)271, Leucine (B)268, and Proline (B)265. A single pi-pi T-shaped bond is formed between Tyrosine (B:393) and the receptor pocket (Fig. 3 B). A pi-sigma bond is also formed with Isoleucine (B:264) which is also attached with an alkyl bond to the ligand. An intriguing observation revealed that most of the bonds in the methamphetamine structure originate from its aromatic ring rather than its linear backbone.

3.2.3. Interactions of MAO-B receptor with Rasagiline, Pargyline and L-theanine

Out of all the bonds observed between these two ligands, hydrogen bonds are generally the strongest. Hydrogen bonds are formed between a hydrogen atom covalently bonded to an electronegative atom (such as nitrogen, oxygen, or fluorine) and another electronegative atom. They are typically stronger than the other interactions listed because they involve significant electrostatic attractions between partial charges. The chemical structure of rasagiline consists of a propargylamine moiety, which is crucial for its inhibitory activity against MAO-B. This propargylamine group undergoes a selective and irreversible binding to the active site of MAO-B, leading to long-lasting inhibition of the enzyme's activity. Having a linear structure, L-theanine bonds with MAO-B receptor with three conventional hydrogen bonds at Arginine 42(A), Glycine 13(A), Glutamic acid 34(A)(Fig. 4 B). No other bonds could be observed within the linear structure of L-theanine. However, hydrogen bonds can prove to be much stronger than most

other interactions. Whereas, rasagiline exhibits only one conventional hydrogen bond at glutamic acid 379(A). There are four weak van der Waal interactions at glutamic acid 303(A), phenylalanine 305(A), valine 381(A), and tyrosine 301(A). A pi-anion bond is additionally observed at glutamic acid 376(A) (Fig. 4 A). It is a type of non-covalent interaction that occurs between a pi system, typically an aromatic ring or a double bond, and an anion. This interaction is characterized by the attraction between the negative charge of the anion and the positive electrostatic potential of the pi system. Lysine 302(A) is attached to the compound with two alkyl bonds. On the other hand, pargyline forms a stacking interaction with Tyr398(B), contributing to stability by engaging the aromatic ring of pargyline with the tyrosine residue of the receptor. A weak hydrogen bond is observed between pargyline and Gly57, which can contribute to ligand stabilization (Fig. 4 C). Several hydrophobic interactions occur between pargyline and Phe343(B), Val294(B), Lys296(B), and Trp388(B), indicating the importance of nonpolar interactions in its receptor binding.

In the interaction study between the three compounds, it is noteworthy that rasagiline does not entirely fit within the pockets of the MAO-B receptor, leaving it surrounded by amino acid residues unlike L-theanine & Pargyline, which interact seamlessly with the receptor from various directions. Rasagiline appears to occupy a single position without fully engaging with the surrounding amino acids. This distinction underscores the differing binding characteristics and interactions of the compounds with the receptor site.

3.3. Free energy binding score of the ligands with the MAO-B receptor.

Free energy binding scores provide an estimate of the strength of interaction between a ligand and its target protein. It represents the thermodynamic stability of the interaction between a ligand (drug candidate) and its target protein (Du et al., 2016). A lower docking score typically corresponds to a higher predicted binding affinity or stronger binding energy (Pantsar & Poso, 2018).

Safinamide displayed the highest binding affinity of -10.3 kcal/mol as compared to the other ones, followed by procarbazine (-8.5 kcal/mol), selegiline (-7.7 kcal/mol), and amphetamine (-6.5 kcal/mol) (Table 1). Interestingly, L-theanine exhibits a higher binding affinity than rasagiline and pargyline. Thus, L-theanine may prove to be an efficient MAO-B inhibitor, whose binding energy with MAO-B receptor is comparable to those of the commonly used inhibitors for the treatment of PD symptoms.

3.4. Comparison of the drug-likeness properties of different ligands.

The drug-likeness study of ligands in a docking study is critical because it evaluates the potential of a compound to become a viable therapeutic drug (Agu et al., 2023). This involves assessing the pharmacokinetic and physicochemical properties of ligands to ensure they align with the characteristics required for successful drug development (Sampat et al., 2022).

Among all the ligands, L-theanine possesses a medium molecular weight, which is neither too high nor too low. The notably higher drug-likeness score of L-theanine (0.24) compared to safinamide (0.08) may be attributed to the exceptionally high molecular weight of safinamide. This may influence its drug-likeness score since molecular weight is a key factor in pharmacokinetics and bioavailability of a compound. Molecular weight of L-theanine is approximately 174.2 g/mol, whereas that of safinamide is 302.3 g/mol. Again, rasagiline and pargyline exhibited lower drug-likeness model scores of 0.13 and 0.04 compared to L-theanine, although their molecular weights are similar to that of L-theanine (Fig. 5). The drug-likeness model score of procarbazine, selegiline, and methamphetamine are 0.90, 0.85 and 0.32 respectively, which are higher than L-theanine.

4. Discussion

Monoamine oxidase-B (MAO-B) inhibitors constitute an important class of therapeutic agents for the management of Parkinson's disease (PD) by reducing dopamine degradation in the central nervous system, thereby improving motor function and decreasing the dependence on levodopa therapy (Finberg & Rabey, 2016; Laban & Saadabadi, 2023; Sekar et al., 2024; Tan et al., 2022; Turnbull et al., 2005). Among the clinically approved MAO-B inhibitors, selegiline, rasagiline, and safinamide have demonstrated efficacy in alleviating PD symptoms, with selegiline reported to be particularly effective when used in combination with levodopa (Binde et al., 2020). Despite these therapeutic benefits, the long-term clinical use of MAO inhibitors remains constrained by several limitations, including adverse drug interactions, irreversible enzyme inhibition, hepatotoxicity, and dietary restrictions associated with tyramine-containing foods, commonly referred to as the "cheese effect" (Chamberlain & Baldwin, 2021; Edinoff et al., 2022; Flockhart, 2012; Pickar et al., 1982; Todorović Vukotić et al., 2021; Van den Eynde et al., 2022). These drawbacks have encouraged the search for safer, naturally derived MAO-B inhibitors.

Green tea and its bioactive constituents have attracted considerable attention because of their neuroprotective and antidepressant properties. Nabavi et al. (2017) reported that green tea catechins exhibit antidepressant-like effects, which may partly arise from their ability to inhibit MAO activity. Similarly, Mazzio et al. (2013) demonstrated that green tea extracts possess significant inhibitory activity against human MAO-B using a high-throughput enzyme microarray assay. Further supporting these observations, Shao et al. (2022) proposed that tea bioactive compounds, including L-theanine, catechins, tea pigments, and GABA, alleviate depression by suppressing hyperactivity of the hypothalamic-pituitary-adrenal (HPA) axis, reducing neuroinflammation, restoring monoaminergic balance, and lowering MAO levels. These findings collectively suggest that tea-derived phytochemicals may exert their neuroprotective effects through modulation of monoamine metabolism.

Previous experimental evidence from our laboratory also indicated that L-theanine significantly improved motor behaviour and restored dopamine levels in an MPTP-induced mouse model of Parkinson's disease (Deb & Borah, 2024). The beneficial effects might be mediated through dopamine receptor agonism or inhibition of dopamine-metabolizing enzymes such as MAO-B or catechol-O-methyltransferase (COMT), although direct evidence was not available. The present molecular docking study provides computational support for this hypothesis by demonstrating that L-theanine interacts favourably with the MAO-B active site.

Our docking analysis revealed that L-theanine exhibited a stronger binding affinity toward MAO-B than the clinically used inhibitors rasagiline and selegiline. Furthermore, L-theanine demonstrated favourable drug-likeness properties, surpassing those of rasagiline, selegiline, and safinamide, indicating its potential suitability as a lead compound for drug development (Fig 6). The enhanced binding affinity may be attributed to favourable molecular interactions within the enzyme active site, including hydrogen bonding and hydrophobic interactions, while its superior drug-likeness may reflect advantageous physicochemical and ADMET characteristics. Although these findings are based on computational analyses and require validation through enzyme inhibition assays and *in vivo* studies, they provide a mechanistic basis for the previously reported neuroprotective effects of L-theanine and support its potential as a safe, natural MAO-B inhibitor for the treatment of Parkinson's disease.

5. Conclusion

Our findings highlight the potential of L-theanine as a natural MAO-B inhibitor with promising neuroprotective implications for Parkinson's disease (PD). Molecular docking revealed a strong binding affinity of L-theanine toward MAO-B, comparable to or exceeding that of established inhibitors such as rasagiline and selegiline, while also demonstrating favourable drug-likeness properties. In addition to MAO-B inhibition, the reported antioxidant, anti-inflammatory, and neurotransmitter-modulating activities of L-theanine further support its therapeutic potential.

Given the limitations associated with currently available MAO-B inhibitors, L-theanine represents a promising natural alternative for the management of PD and other dopamine-related neurodegenerative disorders. However, these *in silico* findings require validation through *in vitro* enzyme assays, *in vivo* studies, and clinical investigations, along with comprehensive evaluation of its pharmacokinetic properties. Overall, this study provides a foundation for the further development of L-theanine as a safe, plant-derived neuroprotective agent for Parkinson's disease.

References

- [1]. Agu, P. C., Afiukwa, C. A., Orji, O. U., Ezech, E. M., Ofoke, I. H., Ogbu, C. O., Ugwuja, E. I., & Aja, P. M. (2023). Molecular docking as a tool for the discovery of molecular targets of nutraceuticals in diseases management. *Scientific Reports*, 13(1), 13398. <https://doi.org/10.1038/S41598-023-40160-2>
- [2]. Baroroh, S.Si., M.Biotek., U., Muscifa, Z. S., Destiarani, W., Rohmatullah, F. G., & Yusuf, M. (2023). Molecular interaction analysis and visualization of protein-ligand docking using Biovia Discovery Studio Visualizer. *Indonesian Journal of Computational Biology (IJCB)*, 2(1), 22–30. <https://doi.org/10.24198/IJCB.V2I1.46322>
- [3]. Binde, C. D., Tvete, I. F., Gåsemyr, J. I., Natvig, B., & Klemp, M. (2020). Comparative effectiveness of dopamine agonists and monoamine oxidase type-B inhibitors for Parkinson's disease: a multiple treatment comparison meta-analysis. *European Journal of Clinical Pharmacology*, 76(12), 1731–1743. <https://doi.org/10.1007/S00228-020-02961-6>
- [4]. Bortolato, M., Chen, K., & Shih, J. C. (2008). Monoamine oxidase inactivation: from pathophysiology to therapeutics. *Advanced Drug Delivery Reviews*, 60(13–14), 1527. <https://doi.org/10.1016/J.ADDR.2008.06.002>
- [5]. Chamberlain, S. R., & Baldwin, D. S. (2021). Monoamine Oxidase Inhibitors (MAOIs) in Psychiatric Practice: How to Use them Safely and Effectively. *CNS Drugs*, 35(7), 703–716. <https://doi.org/10.1007/S40263-021-00832-X>
- [6]. Chew, Z. X., Lim, C. L., Ng, K. Y., Chye, S. M., Ling, A. P. K., & Koh, R. Y. (2023). The Role of Monoamine Oxidase B Inhibitors in the Treatment of Parkinson's Disease - An Update. *CNS & Neurological Disorders Drug Targets*, 22(3), 329–352. <https://doi.org/10.2174/1871527321666211231100255>
- [7]. Cho, H. U., Kim, S., Sim, J., Yang, S., An, H., Nam, M. H., Jang, D. P., & Lee, C. J. (2021). Redefining differential roles of MAO-A in dopamine degradation and MAO-B in tonic GABA synthesis. *Experimental & Molecular Medicine*, 53(7), 1148–1158. <https://doi.org/10.1038/S12276-021-00646-3>
- [8]. Chu, D. C., Okubo, T., Nagato, Y., & Yokogoshi, H. (1999). L-theanine—a unique amino acid of green tea and its relaxation effect in humans. *Trends in Food Science & Technology*, 10(6–7), 199–204. [https://doi.org/10.1016/S0924-2244\(99\)00044-8](https://doi.org/10.1016/S0924-2244(99)00044-8)
- [9]. Deb, S., & Borah, A. (2024). L-theanine, the unique constituent of tea, improves neuronal survivability by curtailing inflammatory responses in MPTP model of Parkinson's disease. *Neurochemistry International*, 179. <https://doi.org/10.1016/J.NEUINT.2024.105830>
- [10]. Deb, S., Dutta, A., Phukan, B. C., Manivasagam, T., Justin Thenmozhi, A., Bhattacharya, P., Paul, R., & Borah, A. (2019). Neuroprotective attributes of L-theanine, a bioactive amino acid of tea, and its potential role in Parkinson's disease therapeutics. *Neurochemistry International*, 129, 104478. <https://doi.org/10.1016/J.NEUINT.2019.104478>

- [11]. Dramard, V., Kern, L., Hofmans, J., Rème, C. A., Nicolas, C. S., Chala, V., & Navarro, C. (2018). Effect of l-theanine tablets in reducing stress-related emotional signs in cats: an open-label field study. *Irish Veterinary Journal*, 71, 21. <https://doi.org/10.1186/s13620-018-0130-4>
- [12]. Du, X., Li, Y., Xia, Y. L., Ai, S. M., Liang, J., Sang, P., Ji, X. L., & Liu, S. Q. (2016). Insights into Protein–Ligand Interactions: Mechanisms, Models, and Methods. *International Journal of Molecular Sciences*, 17(2). <https://doi.org/10.3390/IJMS17020144>
- [13]. Edinoff, A. N., Swinford, C. R., Odisho, A. S., Burroughs, C. R., Stark, C. W., Raslan, W. A., Cornett, E. M., Kaye, A. M., & Kaye, A. D. (2022). Clinically Relevant Drug Interactions with Monoamine Oxidase Inhibitors. *Health Psychology Research*, 10(4), 39576. <https://doi.org/10.52965/001C.39576>
- [14]. Edmondson, D. E., & Binda, C. (2018). Monoamine Oxidases. *Sub-Cellular Biochemistry*, 87, 117–139. https://doi.org/10.1007/978-981-10-7757-9_5
- [15]. Finberg, J. P. M. (2018). Inhibitors of MAO-B and COMT: their effects on brain dopamine levels and uses in Parkinson’s disease. *Journal of Neural Transmission* 2018 126:4, 126(4), 433–448. <https://doi.org/10.1007/S00702-018-1952-7>
- [16]. Finberg, J. P. M., & Rabey, J. M. (2016). Inhibitors of MAO-A and MAO-B in psychiatry and neurology. *Frontiers in Pharmacology*, 7(OCT), 209773. <https://doi.org/10.3389/FPHAR.2016.00340/BIBTEX>
- [17]. Flockhart, D. A. (2012). Dietary restrictions and drug interactions with monoamine oxidase inhibitors: An update. *Journal of Clinical Psychiatry*, 73(SUPPL. 1), 17–24. <https://doi.org/10.4088/JCP.11096SU1C.03>
- [18]. Grabska-Kobylecka, I., Szpakowski, P., Król, A., Książek-Winiarek, D., Kobylecki, A., Głabiński, A., & Nowak, D. (2023). Polyphenols and Their Impact on the Prevention of Neurodegenerative Diseases and Development. *Nutrients*, 15(15). <https://doi.org/10.3390/NU15153454>
- [19]. Harro, J., & Orelund, L. (2016). The role of MAO in personality and drug use. *Progress in Neuro-Psychopharmacology & Biological Psychiatry*, 69, 101–111. <https://doi.org/10.1016/J.PNPBP.2016.02.013>
- [20]. Jaisa-aad, M., Muñoz-Castro, C., Healey, M. A., Hyman, B. T., & Serrano-Pozo, A. (2024). Characterization of monoamine oxidase-B (MAO-B) as a biomarker of reactive astrogliosis in Alzheimer’s disease and related dementias. *Acta Neuropathologica*, 147(1), 66. <https://doi.org/10.1007/S00401-024-02712-2>
- [21]. Jost, W. H. (2022). A critical appraisal of MAO-B inhibitors in the treatment of Parkinson’s disease. *Journal of Neural Transmission (Vienna, Austria : 1996)*, 129(5–6), 723–736. <https://doi.org/10.1007/S00702-022-02465-W>
- [22]. Kuboyama, T., Tohda, C., & Komatsu, K. (2014). Effects of Ashwagandha (roots of *Withaniasomnifera*) on neurodegenerative diseases. *Biological & Pharmaceutical Bulletin*, 37(6), 892–897. <https://doi.org/10.1248/BPB.B14-00022>
- [23]. Kumar, M. J., & Andersen, J. K. (2004). Perspectives on MAO-B in aging and neurological disease: where do we go from here? *Molecular Neurobiology*, 30(1), 77–89. <https://doi.org/10.1385/MN:30:1:077>
- [24]. Laban, T. S., & Saadabadi, A. (2023). *Monoamine Oxidase Inhibitors (MAOI)*. <https://www.ncbi.nlm.nih.gov/books/NBK539848/>
- [25]. Laskowski, R. A., MacArthur, M. W., Moss, D. S., & Thornton, J. M. (1993). PROCHECK: a program to check the stereochemical quality of protein structures. *Journal of Applied Crystallography*, 26(2), 283–291. <https://doi.org/10.1107/S0021889892009944>
- [26]. Marino, B. L. B., de Souza, L. R., Sousa, K. P. A., Ferreira, J. V., Padilha, E. C., da Silva, C. H. T. P., Taft, C. A., & Hage-Melim, L. I. S. (2020). Parkinson’s Disease: A Review from Pathophysiology to Treatment. *Mini Reviews in Medicinal Chemistry*, 20(9), 754–767. <https://doi.org/10.2174/1389557519666191104110908>
- [27]. Mazumder, M. K., Paul, R., & Borah, A. (2013). β -phenethylamine--a phenylalanine derivative in brain-- contributes to oxidative stress by inhibiting mitochondrial complexes and DT-diaphorase: an in silico study. *CNS Neuroscience & Therapeutics*, 19(8), 596–602. <https://doi.org/10.1111/CNS.12113>
- [28]. Mazzio, E., Deiab, S., Park, K., & Soliman, K. F. A. (2013). High throughput screening to identify natural human monoamine oxidase B inhibitors. *Phytotherapy Research : PTR*, 27(6), 818–828. <https://doi.org/10.1002/PTR.4795>

- [29]. Müller, T., & Möhr, J. D. (2019). Pharmacokinetics of monoamine oxidase B inhibitors in Parkinson's disease: current status. *Expert Opinion on Drug Metabolism & Toxicology*, 15(5), 429–435.
<https://doi.org/10.1080/17425255.2019.1607292>
- [30]. Nabavi, S. M., Daglia, M., Braid, N., & Nabavi, S. F. (2017). Natural products, micronutrients, and nutraceuticals for the treatment of depression: A short review. *Nutritional Neuroscience*, 20(3), 180–194.
<https://doi.org/10.1080/1028415X.2015.1103461>
- [31]. Nathan, P. J., Lu, K., Gray, M., & Oliver, C. (2006). The neuropharmacology of L-theanine (N-ethyl-L-glutamine): a possible neuroprotective and cognitive enhancing agent. *Journal of Herbal Pharmacotherapy*, 6(2), 21–30.
<http://www.ncbi.nlm.nih.gov/pubmed/17182482>
- [32]. Pansar, T., & Poso, A. (2018). Binding Affinity via Docking: Fact and Fiction. *Molecules : A Journal of Synthetic Chemistry and Natural Product Chemistry*, 23(8), 1DUMMY. <https://doi.org/10.3390/MOLECULES23081899>
- [33]. Pickar, D., Murphy, D. L., Cohen, R. M., Campbell, I. C., & Lipper, S. (1982). Selective and Nonselective Monoamine Oxidase Inhibitors: Behavioral Disturbances During Their Administration to Depressed Patients. *Archives of General Psychiatry*, 39(5), 535–540. <https://doi.org/10.1001/ARCHPSYC.1982.04290050023006>
- [34]. Regensburger, M., Ip, C. W., Kohl, Z., Schrader, C., Urban, P. P., Kassubek, J., & Jost, W. H. (2023). Clinical benefit of MAO-B and COMT inhibition in Parkinson's disease: practical considerations. *Journal of Neural Transmission (Vienna, Austria : 1996)*, 130(6), 847–861. <https://doi.org/10.1007/S00702-023-02623-8>
- [35]. Riederer, P., & Müller, T. (2018). Monoamine oxidase-B inhibitors in the treatment of Parkinson's disease: clinical-pharmacological aspects. *Journal of Neural Transmission (Vienna, Austria : 1996)*, 125(11), 1751–1757.
<https://doi.org/10.1007/S00702-018-1876-2>
- [36]. Sampat, J Ganesh, Suryawanshi, S. S., Palled, M. S., Patil, A. S., Khanal, P., & Salokhe, A. S. (2022). Drug Likeness Screening and Evaluation of Physicochemical Properties of Selected Medicinal Agents by Computer Aided Drug Design Tools. *Advances in Pharmacology and Pharmacy*, 10(4), 234–246.
<https://doi.org/10.13189/app.2022.100402>
- [37]. Sekar, P. K. C., Thomas, S. M., & Veerabathiran, R. (2024). An overview of the role of monoamine oxidase-B in Parkinson's disease: implications for neurodegeneration and therapy. *Open Exploration 2019 4:4*, 4(4), 308–318.
<https://doi.org/10.37349/ENT.2024.00085>
- [38]. Shao, J., Wei, Y., & Wei, X. (2022). A comprehensive review on bioavailability, safety and antidepressant potential of natural bioactive components from tea. *Food Research International*, 158, 111540.
<https://doi.org/10.1016/J.FOODRES.2022.111540>
- [39]. Szökö, É., Tábi, T., Riederer, P., Vécsei, L., & Magyar, K. (2018). Pharmacological aspects of the neuroprotective effects of irreversible MAO-B inhibitors, selegiline and rasagiline, in Parkinson's disease. *Journal of Neural Transmission (Vienna, Austria : 1996)*, 125(11), 1735–1749. <https://doi.org/10.1007/S00702-018-1853-9>
- [40]. Tan, Y. Y., Jenner, P., & Chen, S. Di. (2022). Monoamine Oxidase-B Inhibitors for the Treatment of Parkinson's Disease: Past, Present, and Future. *Journal of Parkinson's Disease*, 12(2), 477. <https://doi.org/10.3233/JPD-212976>
- [41]. Thomsen, R., & Christensen, M. H. (2006). MolDock: a new technique for high-accuracy molecular docking. *Journal of Medicinal Chemistry*, 49(11), 3315–3321. <https://doi.org/10.1021/JM051197E>
- [42]. Todorović Vukotić, N., Đorđević, J., Pejić, S., Đorđević, N., & Pajović, S. B. (2021). Antidepressants- and antipsychotics-induced hepatotoxicity. *Archives of Toxicology 2021 95:3*, 95(3), 767–789.
<https://doi.org/10.1007/S00204-020-02963-4>
- [43]. Tsuboi, T., Satake, Y., Hiraga, K., Yokoi, K., Hattori, M., Suzuki, M., Hara, K., Ramirez-Zamora, A., Okun, M. S., & Katsuno, M. (2022). Effects of MAO-B inhibitors on non-motor symptoms and quality of life in Parkinson's disease: A systematic review. *Npj Parkinson's Disease 2022 8:1*, 8(1), 1–21. <https://doi.org/10.1038/s41531-022-00339-2>
- [44]. Turnbull, K., Caslake, R., Macleod, A., Ives, N., Stowe, R., & Counsell, C. (2005). Monoamine oxidase B inhibitors for early Parkinson's disease. *The Cochrane Database of Systematic Reviews*, 2005(3), CD004898.
<https://doi.org/10.1002/14651858.CD004898.PUB2>

- [45]. Van den Eynde, V., Gillman, P. K., & Blackwell, B. B. (2022). The Prescriber's Guide to the MAOI Diet—Thinking Through Tyramine Troubles. *Psychopharmacology Bulletin*, 52(2), 73.
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9172554/>
- [46]. Vijiaratnam, N., Simuni, T., Bandmann, O., Morris, H. R., & Foltynie, T. (2021). Progress towards therapies for disease modification in Parkinson's disease. *The Lancet. Neurology*, 20(7), 559–572.
[https://doi.org/10.1016/S1474-4422\(21\)00061-2](https://doi.org/10.1016/S1474-4422(21)00061-2)
- [47]. Xu, Y., Wang, S., Hu, Q., Gao, S., Ma, X., Zhang, W., Shen, Y., Chen, F., Lai, L., & Pei, J. (2018). CavityPlus: a web server for protein cavity detection with pharmacophore modelling, allosteric site identification and covalent ligand binding ability prediction. *Nucleic Acids Research*, 46(W1), W374–W379.
<https://doi.org/10.1093/NAR/GKY380>
- [48]. Yamada, M., & Yasuhara, H. (2004). Clinical Pharmacology of MAO Inhibitors: Safety and Future. *NeuroToxicology*, 25(1–2), 215–221. [https://doi.org/10.1016/S0161-813X\(03\)00097-4](https://doi.org/10.1016/S0161-813X(03)00097-4)
- [49]. Yoto, A., Motoki, M., Murao, S., & Yokogoshi, H. (2012). Effects of L-theanine or caffeine intake on changes in blood pressure under physical and psychological stresses. *Journal of Physiological Anthropology*, 31(1), 28.
<https://doi.org/10.1186/1880-6805-31-28>
- [50]. Zhong, H. J., Liu, L. J., Chong, C. M., Lu, L., Wang, M., Chan, D. S. H., Chan, P. W. H., Lee, S. M. Y., Ma, D. L., & Leung, C. H. (2014). Discovery of a Natural Product-Like iNOS Inhibitor by Molecular Docking with Potential Neuroprotective Effects In Vivo. *PLOS ONE*, 9(4), e92905. <https://doi.org/10.1371/JOURNAL.PONE.0092905>

Statements and declaration

Funding

We express our sincere gratitude for the support and funding provided by the Department of Science and Technology, Govt. of India (DST/INSPIRE Fellowship/2017/IF170756).

Acknowledgements

We gratefully acknowledge Mr. Deep Choudhury for his valuable assistance, as his revisions adorn the figures of this manuscript.

FIGURES AND TABLE:

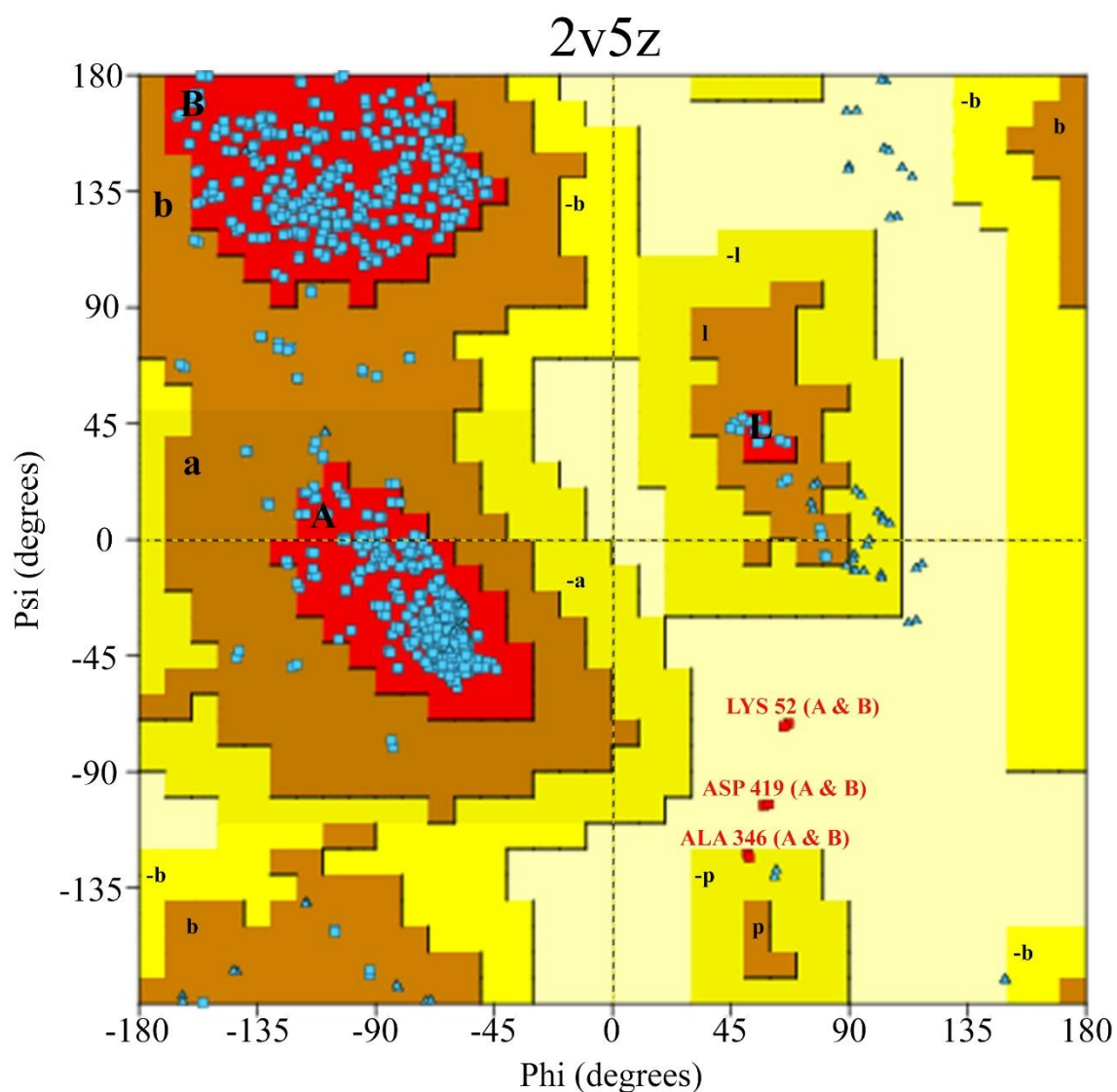


Fig 1. Ramachandran plot analysis indicates that the selected MAO-B receptor (PDB ID: 2V5Z) exhibits excellent stereochemical integrity, with 93.4% of its amino acid residues situated within the most favoured region of the plot. Only 5.9% residues fall in additionally allowed regions. Negligible number of residues fall outside the favoured range. The remaining residues are distributed across additional and generously allowed regions of the Ramachandran plot as generated in Procheck module.

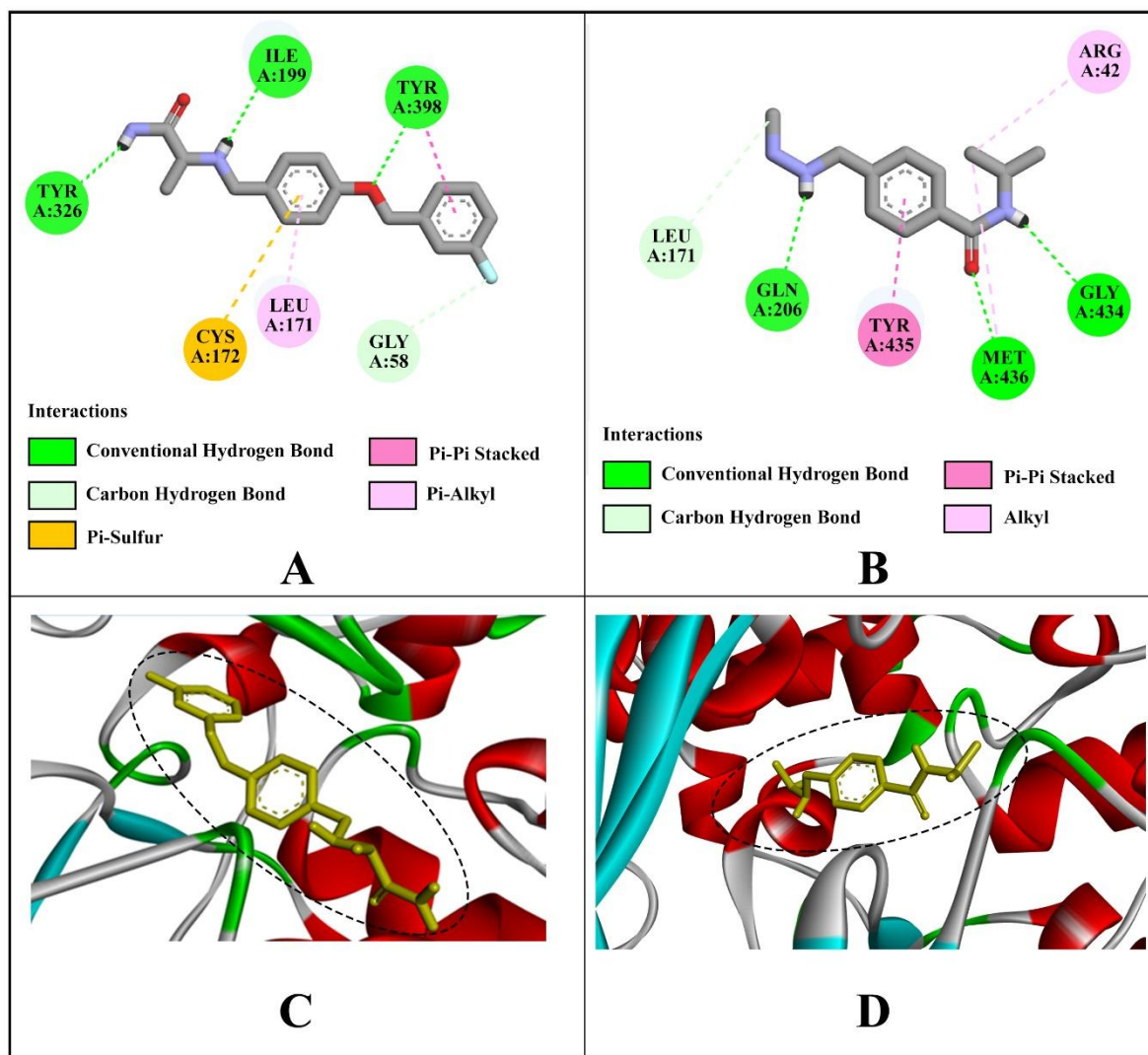


Fig. 2. 2D and 3D view of the close amino acid residues participated in H-bond, van der Waals interaction, and pi-alkyl interaction with phytochemicals within the active pocket of human MAO-B are represented. With (A&C) Sildenafil (B&D) Procarbazine.

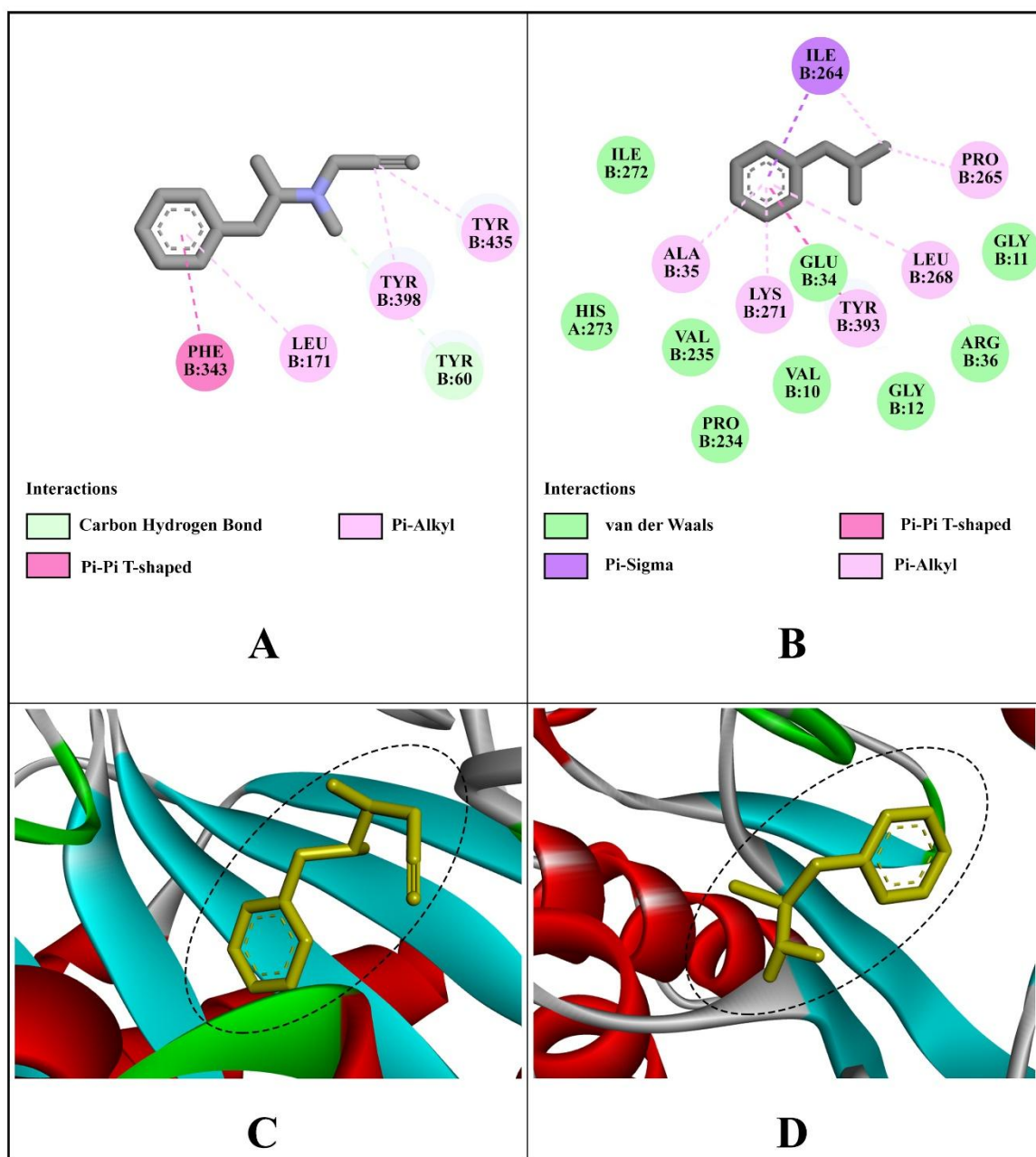


Fig.3.The 2D and 3D representations illustrate the key amino acid residues involved in hydrogen bonding, van der Waals interactions, and Pi-Alkyl interactions with phytochemicals within the active pocket of human MAO-B. Panels (A&C) depict interactions with Selegiline, while panels (B&D) showcase interactions with Amphetamine.

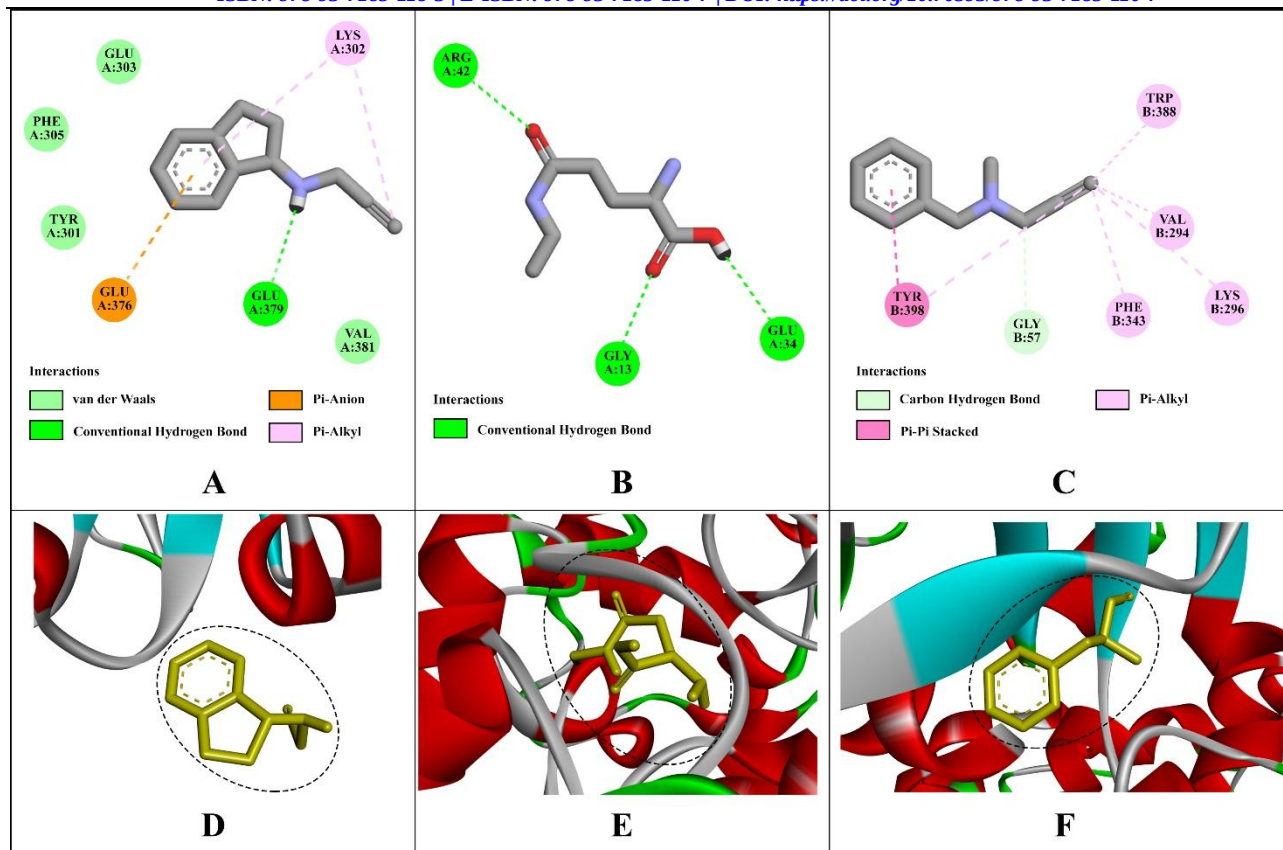


Fig.4. The 2D and 3D representations display the amino acid residues closely involved in hydrogen bonding, van der Waals interactions, and Pi-Alkyl interactions with phytochemicals within the active pocket of human MAO-B. Panels (A&D) illustrate interactions with rasagiline, panels (B&E) depict interactions with L-theanine and panels (C&F) show binding interactions with pargyline.

Table 1. The list of ligands whose interactions were studied with MAO-B receptor along with their docking scores, the various residues involved in their interaction with the receptor.

Ligands	Binding Affinity (kcal/mol)	Interacting residues
Safinamide	-10.3	Tyr A: 326, Ile A:199, Tyr A:398, Cys A: 172, Leu A:171, Gly A: 58
Procarbazine	-8.5	Arg A:42, Leu A:171, Glu A:206, Tyr A:435, Met A:436, Gly A:434

Selegiline	-7.7	Phe B:343, Leu B: 171, Tyr B:398, Tyr B:435, Tyr B: 60
Amphetamine	-6.5	Ile B:264, Ile B:272, Ala B:35, His A:273, Val B:235, Lys B:271, Pro B:234, Glu B:34, Tyr B:393, Val B:10, Gly B:12, Leu B:268, Arg B:36, Pro B: 265, Gly B:11
L-theanine	-6.2	Arg A:42, Gly A:13, Glu A:34
Rasagiline	-5.6	Lys A:302, Glu A:303, Phe A:305, Tyr A:301, Glu A:376, Glu A:379, Val A:381
Pargyline	-4.7	Val B:110, Gly B:12, Leu B:298, Arg B:34, Leu B: 265, Glu B:211

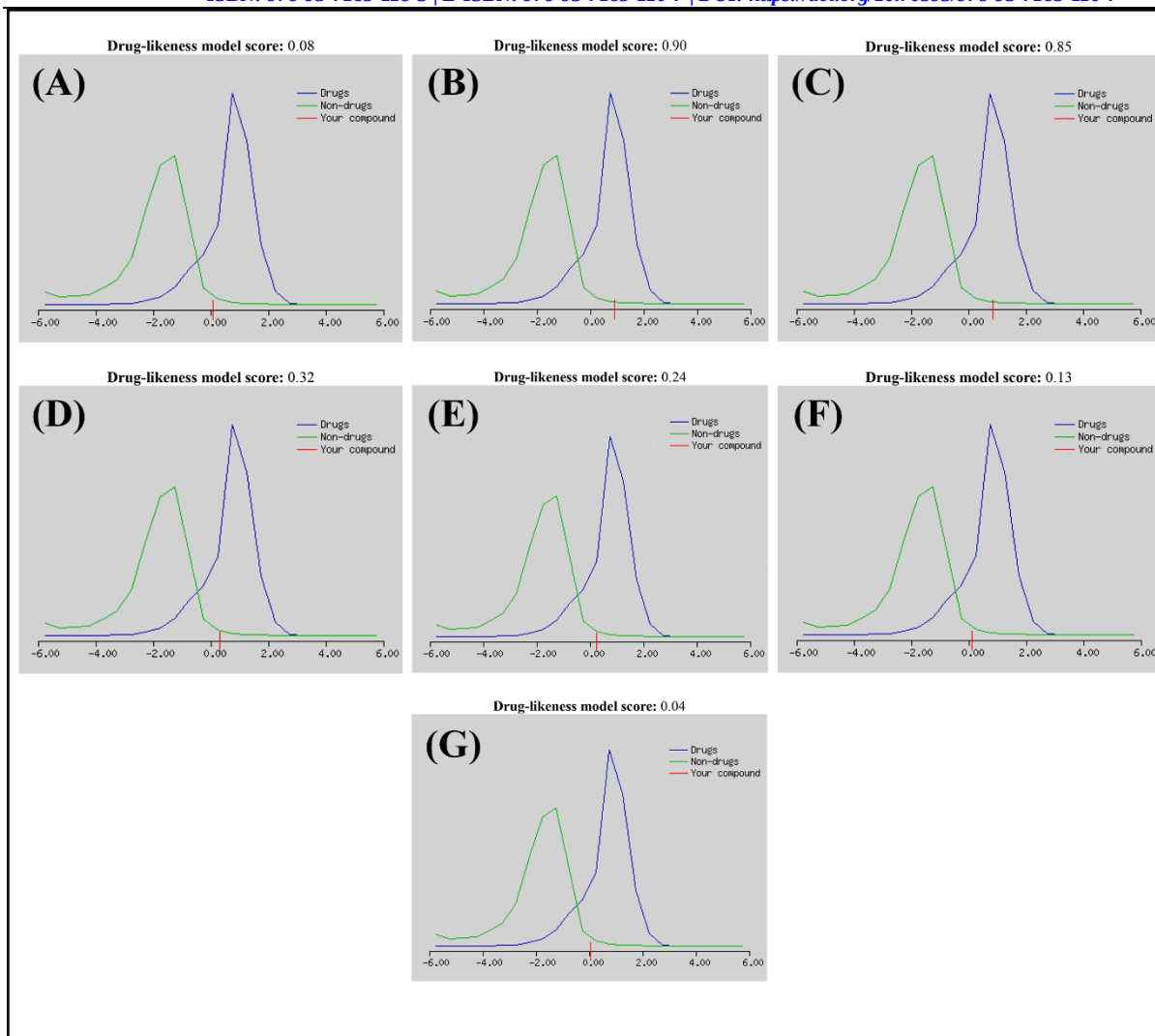


Fig.5. Graphs showing the drug-likeness model score of (A) Safinamide (B) Procarbazine (C) Selegiline (D) Methamphetamine (E) L-theanine (F) Rasagiline (G) Pargyline, as generated through Molsoft LLC.

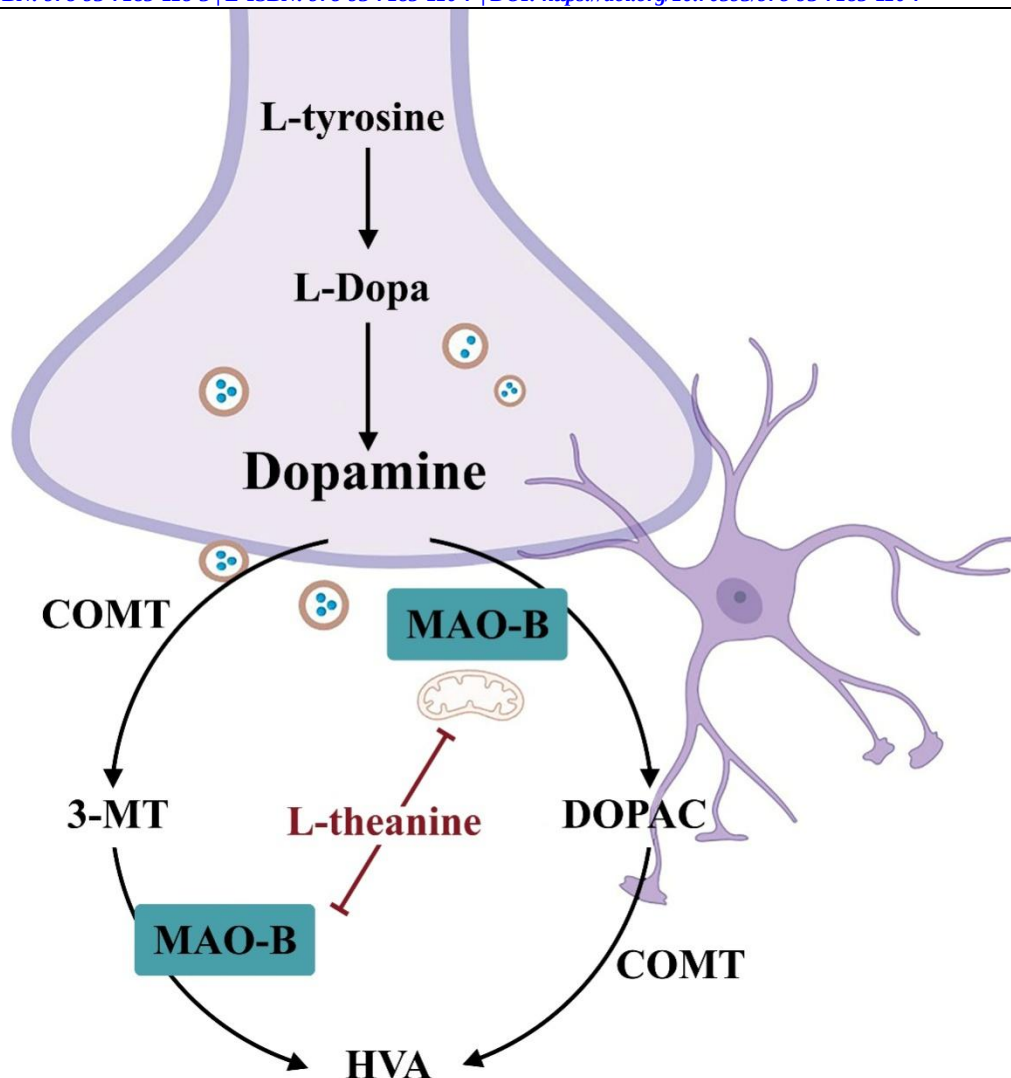


Fig. 6. Increased MAO-B activity leads to excessive dopamine breakdown, reducing its availability and contributing to neuronal dysfunction. In PD, elevated MAO-B activity accelerates dopaminergic neuron loss, further impairing motor and cognitive functions. Existing MAO-B inhibitors, such as selegiline and rasagiline, effectively decline dopamine breakdown but have limitations including potential side effects. L-theanine, a non-protein amino acid in tea, exhibits strong MAO-B binding affinity, surpassing rasagiline and pargyline. Its superior drug-likeness score compared to safinamide, rasagiline, and pargyline suggests its potential as a more effective natural MAO-B inhibitor than the currently available options.